

Exam 3 Review

Exam 3 covers lectures 18-24 on Laplace Transforms

Topics: Laplace Transforms and Inverse Transforms

- Definition

- Properties

Solving IVPs

Step Functions / Dirac Delta

Impulse Response Functions

Convolutions

A formula sheet is provided.

Ex: Find the Laplace Transform of

$$f(t) = \begin{cases} e^t, & 0 \leq t \leq 5 \\ -1, & 5 < t \end{cases}$$

By definition: $\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$

$$= \int_0^5 e^{-st} f(t) dt + \int_5^{\infty} e^{-st} f(t) dt$$

$$= \int_0^5 e^{-st} (e^t) dt + \lim_{N \rightarrow \infty} \int_5^N e^{-st} (-1) dt$$

$$= \int_0^5 e^{(1-s)t} dt + \lim_{N \rightarrow \infty} \left[-\frac{e^{-st}}{-s} \Big|_{t=5}^{t=N} \right]$$

$$= \left[\frac{e^{(1-s)t}}{1-s} \Big|_{t=0}^{t=5} \right] + \lim_{N \rightarrow \infty} \left[\frac{e^{-sN}}{s} - \frac{e^{-5s}}{s} \right]$$

s > 0

$$= \frac{e^{(1-s)5}}{1-s} - \frac{e^{(1-s)(0)}}{1-s} + 0 - \frac{e^{-5s}}{s} \quad (s > 0)$$

$$= \frac{1}{1-s} e^{(1-s)5} - \frac{1}{1-s} - \frac{e^{-5s}}{s}, \quad s > 0, s \neq 1$$

Rewrite $f(t) = \begin{cases} e^t, & 0 \leq t \leq 5 \\ -1, & 5 < t \end{cases}$ using step functions.

$$f(t) = e^t + u(t-5) \left(\underset{\text{on}}{-1} - \underset{\text{off}}{e^t} \right)$$

$$\mathcal{L}\{f(t)u(t-a)\} = e^{-as} \mathcal{L}\{f(t+a)\}$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as} F(s) \quad \text{shift before transform}$$

$$\mathcal{L}\{f(t)\} = \frac{1}{s-1} + e^{-5s} \left(\mathcal{L}\{-1 - e^t \mid_{t \rightarrow t+5} \} \right)$$

$$= \frac{1}{s-1} + e^{-5s} \left(\mathcal{L}\{-1 - e^{t+5}\} \right)$$

$$= \frac{1}{s-1} + e^{-5s} \left(-\frac{1}{s} - \frac{1}{s-1} e^5 \right)$$

$$= -\frac{1}{1-s} - \frac{e^{-5s}}{s} + \frac{e^{-5s} e^5}{1-s} \leftarrow e^{(1-s)5}$$

Ex: Solve $y' - 2 \int_0^t y(v) \sinh(t-v) dv = 1$, $y(0) = -1$

$\Rightarrow (sY(s) - \cancel{y(0)}) - 2 \left(\frac{1}{s^2+1} \cdot Y(s) \right) = \frac{1}{s}$

$\sinh(t) * y$

$\mathcal{L}\{f * g\} = F(s)G(s)$

$\Rightarrow Y(s) \left(s - \frac{2}{s^2+1} \right) + 1 = \frac{1}{s}$

$\Rightarrow Y(s) \left(\frac{s(s^2+1) - 2}{s^2+1} \right) = \frac{1}{s} - 1 = \frac{1-s}{s}$

$\Rightarrow Y(s) = \frac{1-s}{s} \cdot \frac{s^2+1}{s^3+s-2} = \frac{(s^2+1)(1-s)}{s(s^3+s-2)}$

s^3+s-2

rational roots: $\pm 1, \pm 2$ so $s-1$ is a factor of s^3+s-2

$s=1: 1+1-2=0 \checkmark$

$s-1 \overline{) s^3+s-2}$

$-(s^3-s^2)$

$\hline s^2+s$

$-(s^2-s)$

$+2s-2$

$-(+2s-2)$

$\hline 0$

$\Rightarrow s^3+s-2 = (s-1)(s^2+s+2)$

$$Y(s) = - \frac{(s^2+1)(1-s)}{s(s-1)(s^2+s+2)} \rightarrow -(s-1)$$

$$\Rightarrow Y(s) = - \frac{s^2+1}{s(s^2+s+2)} \quad \nearrow \quad \frac{1}{s} \cdot \frac{-s^2+1}{s^2+s+2}$$

$$\frac{-(s^2+1)}{s(s^2+s+2)} \stackrel{\text{PED}}{=} \frac{A}{s} + \frac{B(s+\frac{1}{2}) + C \cdot \frac{\sqrt{7}}{2}}{(s+\frac{1}{2})^2 + \frac{7}{4}}$$

$$\boxed{s((s+\frac{1}{2})^2 + \frac{7}{4})} \Rightarrow -(s^2+1) = A[(s+\frac{1}{2})^2 + \frac{7}{4}] + s[B(s+\frac{1}{2}) + C \cdot \frac{\sqrt{7}}{2}]$$

$\uparrow$ $\uparrow$
 $(s^2+s+\frac{1}{4})+2-\frac{1}{4}$

check values to find A, B, C : $A = -\frac{1}{2}$, $B = -\frac{1}{2}$

$$C = \frac{3}{2\sqrt{7}}$$

$$\text{So } Y(s) = \frac{-1/2}{s} + \frac{-\frac{1}{2}(s+\frac{1}{2})}{(s+\frac{1}{2})^2 + \frac{7}{4}} + \frac{\frac{3}{2\sqrt{7}} \cdot \frac{\sqrt{7}}{2}}{(s+\frac{1}{2})^2 + \frac{7}{4}}$$

$$\Rightarrow y(t) = -\frac{1}{2} - \frac{1}{2} e^{-1/2 t} \cos(\frac{\sqrt{7}}{2} t) + \frac{3}{2\sqrt{7}} e^{-1/2 t} \sin(\frac{\sqrt{7}}{2} t)$$

Ex: Given $y'' - 5y' + 6y = g(t)$ if $y(0) = 0$
 $y'(0) = 0$

then $y(t) = h(t) * g(t)$

$h(t)$ impulse response function

$$= \mathcal{L}^{-1}\{H(s)\}$$

transfer function

$$H(s) = \frac{1}{s^2 - 5s + 6} = \frac{1}{(s-2)(s-3)}$$

$$H(s) = \frac{1}{s^2 - 5s + 6}$$

$$0 < s < \infty$$

$$\text{So } h(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-2)(s-3)}\right\} = e^{2t} * e^{3t}$$

$$\frac{1}{s-2} \cdot \frac{1}{s-3} \rightarrow e^{2t}, e^{3t}$$

$$= \int_0^t e^{2(t-v)} e^{3v} dv = \int_0^t e^{2t+v} dv$$

$$= \int_0^t e^{2t} e^v dv$$

$$= e^{2t} e^v \Big|_{v=0}^{v=t} = e^{2t} e^t - e^{2t} e^0 = e^{3t} - e^{2t}$$

$$\text{Therefore } y(t) = (e^{3t} - e^{2t}) * g(t)$$

Ex: Suppose $g(t) = \delta(t-1)$

$$\text{then } y(t) = (e^{3t} - e^{2t}) * \underset{\substack{\uparrow \\ a=1}}{\delta(t-1)}$$

Recall: $f(t) * \delta(t-a) = u(t-a)f(t-a)$

$$\Rightarrow y(t) = u(t-1) [e^{3(t-1)} - e^{2(t-1)}]$$

$$\text{Ex: } \mathcal{L}\{t^3 \delta(t-4)\} \rightarrow \mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

$$= (-1)^3 \frac{d^3}{ds^3} \mathcal{L}\{\delta(t-4)\} = -1 \frac{d^3}{ds^3} (e^{-4s})$$

$$= -1(-4 \cdot -4 \cdot -4 e^{-4s})$$

$$= 64 e^{-4s}$$

$$\text{or } \mathcal{L}\{t^3 \delta(t-4)\} \overset{a=4}{=} 4^3 e^{-4s} = 64 e^{-4s}$$

$$\mathcal{L}\{f(t) \delta(t-a)\} = f(a) e^{-as}$$

$$\text{Ex: } \mathcal{L}\left\{\int_0^t (t-v) \sinh(\underset{\substack{\uparrow \\ b}}{3v}) dv\right\} = \mathcal{L}\{t * \sinh(\underset{\substack{\uparrow \\ b}}{3t})\}$$

$$= \frac{1}{s^2} \cdot \frac{\overset{3 \leftarrow}{3}}{s^2 + 9} = \frac{3}{s^2(s^2 + 9)}$$

$$\text{Ex: } \mathcal{L}^{-1} \left\{ e^{-4s} \frac{s}{s^2+4} \right\}$$

$a=4$

$$\mathcal{L}^{-1} \{ e^{-as} F(s) \} = u(t-a) f(t-a)$$

↑
shift after
inverse transform

$$\mathcal{L}^{-1} \left\{ e^{-4s} \frac{s}{s^2+4} \right\} = u(t-4) [\cos(2t) |_{t \rightarrow t-4}]$$

$$= u(t-4) \cos(2(t-4))$$

Ex: Evaluate $\int_{-\infty}^{\infty} \underbrace{e^{2t} \cos(3t)}_{f(t)} \delta(t-\pi) dt$

$a=\pi$

$$\int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a)$$

$$= e^{2t} \cos(3t) \Big|_{t=\pi}$$

$$= e^{2\pi} \cos(3\pi)$$

$$= \underline{-e^{2\pi}}$$

$$ay'' + by' + cy = g(t) \quad y(0) = y_0, \quad y'(0) = y_1$$

$$y = y_h + y_p = y_h + (h * g)$$

$$y'' - 5y' + 6y = g(t), \quad y(0) = 1, \quad y'(0) = -1.$$

$$y_h = c_1 e^{3t} + c_2 e^{2t}, \quad 1 = y(0) = c_1 + c_2$$

$$y' = 3c_1 e^{3t} + 2c_2 e^{2t}, \quad -1 = y'(0) = 3c_1 + 2c_2$$

$$\Rightarrow c_1 = -3, \quad c_2 = 4$$

$$\text{So } y_g = (1 - 3e^{3t} + 4e^{2t}) + (e^{3t} - e^{2t}) * g(t)$$

$$\text{Note: } (h * g)(0) = 0 \quad \int_0^t \underbrace{e^{t-\tau}}_{=1} d\tau = 0$$

$$(h * g)'(0) = 0$$